Pixel

Hi everyone, hopefully you can hear me okay.

Thanks for that wonderful introduction.

I'm Awais, I'm the founder and CEO of Pixel Space Technologies.

As Minoo was mentioning, we started this company now 7 years ago.

When the company started, I was still pursuing undergraduate studies at Bitsplan, so I had not graduated yet.

It was my first, and hopefully the only, job that I did for the rest of my life.

There were a couple of very quick experiences at the university that can still be led to Pixel, very briefly.

The first was being part of a student satellite team, right?

As someone who has been fascinated with space, when you enter college and you find out that there's a team that is working with a scientist that is, well, learning how to design, build, and operate satellites, the first step for me was to get into that team and learn how to sort of do that myself.

And the first and second year of university was spent doing a lot of tinkering at this lab, the student satellite team, rather than going to the classes.

And then, thankfully, there was this new opportunity around the Hyperloop that came in, which was around 2017, when Elon Musk was fascinated, or obsessed, with the concept of the Hyperloop.

And SpaceX decided to conduct a competition around it.

They built this one-mile-long vacuum tube at their headquarters in Los Angeles and asked for teams from around the world to come and race the fastest Hyperloop pod.

Basically, this is a vehicle that's supposed to go within a vacuum tube to counter the effects of the air friction that would have been running.

We decided to honestly participate because we thought it would be a good chance to get to meet Elon Musk, nothing beyond that.

But to our very pleasant surprise, we got through the first stage, and then after multiple evaluation stages by the SpaceX engineering team, we were selected as one of the 20 finalist teams.

Now, that experience taught me personally quite a bit on how you actually build hardware that has not been built before, how you work with engineers at the cutting edge of space technology, and we had the back-and-forth with the SpaceX engineering team.

And really, how do we figure out to put a Hyperloop pod together, as college students, take it all the way to Los Angeles?

We had come down to Bangalore for a duration of a couple of months during our summer vacation to sort of make this happen.

We begged, borrowed, did steal a lot of things to work with folks to put this together, from aerospace companies like HAL, to electronics companies, to carbon fiber, composite companies.

We were able to build it, take it all the way to the SpaceX headquarters in Los Angeles, and directly present to Elon Musk himself and the rest of the SpaceX team there.

So personally, that was the eureka moment to look back in August 2017 and decide that space is going to be the space for me for the rest of my life.

So it was a very easy calling.

Working at NASA or SpaceX was not an option because they only hire U.S.

citizens or permanent residents.

Working at this looked seemed like a very long way to get to a point where impact could be made.

But the Hyperloop experience had given that confidence that, you know, if a bunch of young, passionate folks put our minds together and want to work on something, you can really put it together as well.

And so we said, okay, what's the next logical step?

Let's build and launch some satellites.

So coming back from the Hyperloop experience, entering the final year of college, I looked at what was happening in the domain of space technology.

You could build rockets.

Rockets take your equipment from point A to point B.

Point A being on ground, point B being in space.

You could try and build a manufacturing setup in space, where you could do pharmaceuticals or specific types of materials, but that was way far out in the future in terms of, I think, more sci-fi than science fact.

But as college students, without access to any money, what is it something that we could do?

And with the experimentation that we were doing, we narrowed down on identifying that there's a lot of satellite imagery that comes down from space, right?

And this is one dataset that actually gives you a bird's-eye view of what's happening around the planet.

You want to know what's happening with the melting polar ice caps up in the Arctic or the Antarctic?

You can use satellite imagery.

You want to know what's happening all the way in the Amazon rainforest in terms of its deforestation or the amount of carbon it is absorbing?

You can do that with satellite imagery.

So it seemed logical that there could be a lot of impact that could be done if you took this massive amount of data that's coming down from space and used artificial intelligence models to make sense of the imagery that's coming, to parse through it, and then provide the insights and the inputs so that organizations can become more efficient or save money, governments can become more eff icient and take right policy decisions, and in general we can take better care of our planet.

So that's what we started with, which was the vision of building a health monitor for our planet.

See what's happening around the globe, see how these phenomena is changing or moving, and most importantly, be able to predict what is to come.

Be able to predict that if the deforestation rate of the Amazon is increasing or decreasing, how will that affect the concentration of the greenhouse gases, for example, or if the rate of the polar ice caps melting increases, how will it affect the health of the coral reefs around the globe, and things like that.

So the idea was fairly straightforward.

There's this dataset that's coming down, just happens to come down from space.

There's AI models, and this was pre-ChatGPT days, all the way back in 2019, computer vision algorithms and deep learning algorithms that we sort of had to use here.

But it seemed like the intersection of a problem that was exciting, that many people are passionate about, it seemed like something that could really be impactful also for the planet.

And if you're able to really pull this off and you have a digital twin of the planet that is telling us what is going to happen, we can all take better care of it collectively, right?

But we realized when we started that there was a little bit of a problem in the sense that there are certain issues and problems that we should be seeing, but we couldn't.

A few examples are here on the screen.

If you wanted to identify the fact that half a trillion dollars' worth of crops every year are lost just because we can't detect crop diseases or pest infestations or nutrient deficiencies in time around the globe, or the fact that oil and gas leaks from pipelines that are around the globe leads to not only financial losses for the companies but obviously is bad for the planet.

And last but not least, all the pollutant sand that we see, whether it's in the water body, whether it's in the air, or whether it's in the soil where we grow our food.

There was a conjunction here of things that if you could help see, would help save money for certain organizations, and at the same time help make the planet better.

And so if you can't see the problems, then you can't solve it.

So what was the next logical step?

It's to be able to build a better kind of satellite that could then see these problems.

I could take a very crude analogy in a sense here.

Let's say I get into an accident, I have a broken bone, and go to a hospital.

The doctor can't use a phone camera to see what's broken.

The doctor would need to use an X-ray machine, which is a differently kind of built kind of hardware, to be able to permeate what is unseen to see the broken bones.

Or a similar analogy with an MRI machine or an ultrasound scanner.

So similarly, if existing satellites with their cameras up there couldn't see these, then we had to build and deploy a better kind of satellites.

You could see something with these satellites up there, obviously.

We have seen images on Google Earth.

You could see the fact that there's what's the basic crop health level of our crops.

Visually, you could identify where there's pipelines or oil and gas fields.

You could look at where's water, where's land.

But you really needed to go way beyond that to identify the fact that there's specific pest infestations and crop diseases cropping up here and how they are spreading, or what nutrient deficiencies are lacking in this part of the area versus not.

You wanted to go beyond just looking at oil and gas pipelines to actually see if there is a methane leak there.

It's an invisible methane leak.

Where has it begun?

How big is it?

Where is it moving towards?

You should not only be identifying basic maps of, here's a water body, here's land, here's a farm, but actually what is contaminating that water body?

Where is it coming from?

What chemical is it?

And what side effects could it have?

And to be able to go from these visual aspects to being able to see these unseen things is where the gap was.

And the way to do that was to build and deploy our own constellation of satellites that we call hyperspectral satellites.

In a very crude sense, human eyes can only see things in three wavelengths: red, green, and blue.

When we see a rainbow, it's basically different shades of red, green, and blue.

All the pixels on the screen that you have on your phones or your laptops or on the screen behind me are all a combination of red, green, and blue pixels, right?

So there are three wavelengths.

Hyperspectral cameras or hyperspectral satellites that we put up there see things in 150 wavelengths of light today.

It'll be 300 wavelengths of light next year.

And that is really what makes us see the unseen.

And the constellation is designed in a way such that we can provide global coverage and we can do it daily, which means that if you're looking at Mumbai today, we can come back and look at it again in 24 hours.

And we can do that not only for Mumbai, we can do that for New York, we can do that for Tokyo, we can do that for Norway, New Zealand, anywhere on the globe.

So global coverage with a daily revisit or a daily access.

But it's really the fact that you have a hyperspectral imager that sees things in 50 times richer detail.

That then allows us to take it and put it into your AI framework.

And here is really where AI becomes very powerful.

Humans, it's not possible for us to go beyond just the visible range and zoom in and try to identify.

You are looking at various different chemical signatures here.

You are looking at an enormous amount of data.

Since it's 150 bands compared to 3 bands, it's literally that much of an amount of data, which is not possible for humans to be able to do it at the scale that we are capturing.

And that's really where artificial intelligence truly matters to be able to take these large datasets and start to make sense from them.

But it is really the camera that is built behind it, the hardware, that enables that analysis to be unique, because what you're capturing as a dataset is unique.

You own a unique dataset, and the insights, the unique insights that come out of it are that much more powerful, right?

So we started by building our own hyperspectral cameras because that was really the genesis of all this as I just talked about.

This took quite a few years for our team to iterate.

We started with a demo satellite all the way back in 2021, which became India's first-ever private imaging satellite outside of ISRO, right?

ISRO has been doing a wonderful job for the last 60, 70 years.

But to be able to do it privately without any support from there, we began in 2019, and by 2021, we had the first experimental satellite up there.

We followed it up with two more quick iterations that made the resolution better.

And then we put this camera on a satellite platform that we built, right?

And the satellite platform has to be able to sustain this camera up there autonomously.

You are not in touch with this satellite 24/7.

You're only in touch with the satellite once it comes over a ground station, and that happens only every 19 minutes or every 120 minutes.

And so what is the satellite then doing for the rest of the 80 minutes out of the 90 minutes that you are not in touch with it?

It has to be truly autonomous for it to figure out which way it needs to look for its image, and then when it comes over a ground station, how it needs to orient itself so that the antenna points toward the ground station and is beaming it down.

So a lot of artificial intelligence, autonomous systems become very key for the satellites to work by themselves when you're not in touch with them, interact between each other to figure out which satellite is best suited to take an image.

When we send a command up to the constellation, we just say, "Use me to take an image of this point on Earth," and then the constellation and the intelligence figures out which is the satellite that's closest, which one needs to image it, which ground station needs to be used, and all of it makes sure that the cameras can point where they're pointing, they can image,

and then they can downlink that data.

These are the six commercial-grade Firefly satellites that we launched last year.

So we started with those three demonstration satellites in '21 and '22.

But in 2025, we ended up launching these Firefly satellites, which was India's first private satellite constellation to ever launch, and also became the world's highest resolution hyperspectral satellite constellation.

You can see the satellites being deployed there by the SpaceX rockets over two launches that we did there.

Once they're deployed into the orbit, the satellites kind of have to take care of themselves and will be able to communicate with them and tell them what to do.

These are some of the very colorful hyperspectral images that came down from those very same satellites that we built and launched.

But while they look colorful and beautiful to look at, it is really what you're able to extract from them that makes it possible for the customers to use it and why this becomes useful, right?

And that's where your AI models, that's where your algorithms really come into the picture to make all of these use cases possible.

Because at the end, it's the same hardware.

It's the same camera.

The satellites are not changing.

Once you have launched them up there, you can't go up there and make changes, at least not yet.

But then once the images come down over a certain area, what you're able to extract from that depends on the model that you're using.

For example, the same image could have an agricultural farm on the top left.

It could have an oil and gas field on the bottom left.

It could have something else in the rest of the algorithm.

But when you run, let's say, a water quality model on the image, it will look at wherever there is water and tell you what is present in that water body.

Or you don't want to know what's in the water, you want to know what's happening with the crops.

You can run a crop health model that's present, and then it'll tell you what's happening with the crops, and then it'll forget the rest.

Or you could combine it and run multiple models to give you a 360-degree view of the fact that what's happening with the farms and with the water and with the soil and with something else.

So everything from agriculture to forestry to environment, this dataset helps us look at everything from nutrient content of the soil across different entire countries and entire continents.

We can identify species and subspecies classifications within different forests and how they are changing.

We can identify risks of forest fires.

What is the moisture level of certain trees somewhere?

Where will the forest fire likely begin?

And if it begins here, where is it likely to spread?

So the resources can be better planned for when such an eventuality actually happens.

And there's a lot of interest these days around carbon credits.

But who's actually monitoring and verifying what is the actual amount of carbon credits present there?

That happens once you identify what tree you're looking at, what species it is, what is the above-ground biomass, etc.

And then we talked about the water quality as well.

Beyond that, for the natural resources sector, whether it's mining or whether it's oil and gas, we can help detect where the.

might be, you know, everyone, every country is looking for critical minerals these days to identify what might be necessary for the electric vehicle transition that we are looking at, whether it's neodymium, cobalt, lithium and others.

Hyperspectral data is unique because the spectral signatures you detect can help identify all of that.

You run a model, it'll tell you where exactly to look for a specific mineral and therefore mining companies, the governments can focus on those areas to be able to go and extract those.

for oil and gas pipelines and the refineries making sure that as and when there are leaks we go and plug those without it becoming a larger problem.

We've all at some point read about the Horizon Well BP leak that happened and that was detected earlier that there were satellites keeping a track of that and then there's obviously the government sector for everything from national security applications identifying what's happening at the borders beyond the borders are there bad actors that are testing some kind of chemical and biological weapons are there troops that are being amassed near a particular border?

This is something that was important for Ukraine to know, for example, when Russia was mobilizing the forces for them to be sort of prepared then.

And then disaster preparedness and response.

We've all heard the terrible news that happened in Nepal-Tibet border, but how do you identify what's actually happening and what the risk is?

How do you quantify satellite imagery?

You can see things that is very difficult for humans to do.

And with our hyperspectral imagery combined with artificial intelligence models, you can actually go ahead and Start creating these predictive models, and when these things happen, it can give you a good sense of what has actually happened as well.

Now, how do the customers actually use it?

Right, you have satellites up there.

customers are directly interfacing with them, we have imagery coming down, but imagery isn't really a tangible asset to work with, especially at perspective, because it's not like your usual image that you can zoom in and see things.

So we have a platform called Aurora, this was built specifically to be an AI-enabled geospatial analytics platform, where we wanted to remove the entire headache or the entire troubles from a customer standpoint that of the uniqueness data, we've identified that the geospatial works for us, but how do we really get to the insight?

And that's where a platform that enables someone to come in and interface it like on Google Earth.

You can go in, you can type the area of interest.

For example, let's say you want to know what's happening with the lake in the Greater Bombay region.

You can use that as an area of interest.

You can run the water quality model.

So in this example here, we are looking at the city of Munich and we're comparing the vegetation changes that have occurred over different months in a certain year, right?

And you can see that whether it's due to rainfall, whether it's due to something else, there has been a significant shift in the amount of vegetation in the city.

And you can do that at whatever scale you want, you can do it in whichever country you want, you can do it for whatever use case you want.

And it's not only limited to vegetation, you can do it for water quality, you can do it for, you know, burn ratio index, which is your forest fires.

And the solution here basically was to remove the complexity from the customer, whether they want to just download any of our pre-built, pre-trained, pre-verified models, or use any of those to integrate with existing workflows.

They don't necessarily even have to work with the front end that we have, they could integrate it at the back end with existing workflows.

Let's say for example, we're working with Tata in the Northeast for their tea plantations, and they have an internal workflow that tells them what's happening there.

They don't have to use the Aurora platform as a front end, you could just integrate the back end with the same platform that then tells them what is going what is going wrong, alerting them to something that needs to be done to take better care of those plantations.

And also now with the LLM revolution that has happened, it's as easy as coming to the platform, ask it in plain English what you want to do.

Tell me about how the health status of the lakes has changed in the Bombay city over the last one year.

The platform will figure out everything as you ask it for the last 12 months.

It will figure out to take one image of this region over the last 12 months on a monthly basis.

It will run the water quality model and will generate the report for you.

And then you can interact with it very interactively to ask it to change certain things or present it to you in a different way or even ask questions as to, okay, it looks colorful, but what am I actually seeing here?

But that's really where AI has become a force multiplier for a sector like ours, where you don't need to now have any complexity, you don't need to have a team of data scientists, you don't need to have learned many years of remote sensing, you can just come in, ask it in plain English, makes it accessible to anyone and everyone to come in and say, okay,

I want to see what's happening in this area, tell me what's happening, the model will do it for you.

The other thing that we are expanding into is, you know, since we're building satellites for ourselves, the government of India kind of asked us, would you want to build satellites for this as well?

And so we said, you know, everyone keeps talking about SaaS, let's do SaaS as well.

But in that case, it's satellite as a service, where we provide the entire satellite as a product as well.

Here is where, you know, it goes from planetary intelligence that we're doing to planetary infrastructure.

That then leads to the intelligence that gives insights here, right?

So, whether it's working with the Indian Air Force for building an interconnected set of satellites that work autonomously to provide a deterrent mechanism to identify what's happening at the borders, beyond the borders, but also with the Department of Space to integrate with their existing systems.

it becomes not just provide data and analytics now, actually build and provide the hardware that enables that intelligence for entire countries and entire nations.

And that's why I use the word sovereignty here.

Every country now wants its own assets that it can control, that it can see that we have not only sovereign hardware, but we have sovereign data that's coming out of that.

Which is an extension of the same conversation we keep having around foundational models, right?

Owning our own sovereign models gives us that much more power to be able to make sure that no one else can cut off access.

It's a very similar thing with the data and analytics that come out of it here as well.

So what that has led us to become now is a vertically integrated planetary infrastructure company that provides intelligence for a changing planet, whether it's our own fleet of satellites that capture images and provide the data, whether it's the analytics that comes out of that by fusing multiple data sets by running the analysis using AI algorithms,

but also finally the entire satellites themselves that interconnect with IoT devices on ground or other communication satellites to provide

interconnected layer for our planet to better understand itself.

So that's about Pixel and how we are using artificial intelligence.

The theme of the conference is AI versus human.

I think in our case, as you see, it's AI and human that we will continue to work and make sure that we can continue to take much better care of this one planet that we call home.

Thank you.

Thanks, thanks, Abhishek.

So, you know, there are a bunch of questions.

So, I'm going to just ask you two, three buckets of it.

Firstly, there is a lot of operational questions, more about how real time is the analysis.

Question primarily was centered around the Nepal tragedy, is whether that is something that could have been picked up on time.

Does it take a lot of time to process the images?

Yeah, so there's two parts to that question, I think.

Number one is, if you have And we have captured an image, let's say we had captured an image of when the cloud burst happened, the glacier burst happened, the analysis itself doesn't take a lot of time, one because we have actual NVIDIA hardware on the satellites that does edge computing.

you can actually run models on the satellite itself.

And even if you're not doing that on edge, it takes about 15 minutes at the best case scenario for the image to come down and the analysis to happen.

But the important question is, were there satellites that were taking those images and had someone told us to take those images?

Because it's not possible for satellites to be everywhere all the time, unlike CCTVs, because we have very precious few assets up there and they're constantly orbiting.

So the answer is, If for whatever reason there was an image that was taken at the moment it happened, there could have been a possibility that you could have alerted it.

The analysis can be fast, but the question is what happens?

You know, from the operational part to more in terms of conceptualizing the entire revenue pools.

So India, you're of course, you're not limited to India, right?

You could very well do P, you can globalize well.

So how do you actually look at your own revenue model, how many satellites have been required?

More importantly, is it a push model that you could really create and use for it and then go around asking clients or corporates to basically panel with you?

So how does the entire process work?

If there's a time number that you could share as well would be great.

Yeah, so it has been a global business from day one.

When you look at India, India has had a wonderful supply chain ecosystem thanks to ISRO.

But there hasn't been a commercial ecosystem for buying because ISRO used to build satellites in the government agency.

the other ministries used to use that data which are also government agencies and so it was kind of internal.

So there was never really a market that was developed here in terms of companies that were buying satellite data.

So that's something that we had to create here, which is why, but once you send the satellites up into orbit, they're orbiting around the globe.

There's no limits to where you can capture it.

It's global access every day.

So the United States is the largest market because they have been buying satellite data for many decades now from commercial entities.

Europe is the second largest market followed by Asia, right?

Global from day one, that is how it makes sense for the ROI as a satellite is spending quite a few millions of dollars for each satellite and launching constellation it adds up.

how do you make sure that they are the most productive when they are up there?

You take images everywhere that they are going in every orbit over every geography.

And so we work with NASA, for example, as mentioned earlier for environmental sciences.

We work with Ministry of Agriculture in India to identify what's happening here.

We work with organizations like Corteva in Europe where they're using it for identifying what's happening in their farms, with Rio Tinto in Australia, in Africa, and other places that they have.

So it is global.

The TAM number is roughly about two aspects to it.

There's just a data purchase which is about $4.5 billion a year for satellite imagery purchases that happen.

And there's about 14 billion of analytics that is built on top of different kinds of satellite imagery.

So it's those two that we look at in terms of TAM for data and analytics.

So if you could share, if not your company, India specific perhaps, where are Indian companies today on the overall revenue pool?

If you could share yours, it'll be great.

How do you use the journey?

Because I guess you wanted to resolve the broader element of the technology part.

Now, I think it's more demand and you keep shooting more satellites.

Yeah, so probably a pull and push question a little bit as well, right?

So I think globally in certain industries and geographies, the early stages have been thankfully a little bit of pull because Companies already buy satellite data, they have budgets for satellite data.

We're providing a new data set that does a lot more.

So we don't have to teach them why satellite data, we just have to go and show them how our satellite data can help them see 50 times more than existing data so that becomes cool.

But that's a very limited, early sophisticated customer base.

So the larger customer base, let's say, They have not traditionally used a lot of satellite data, so it's a push thing where you have to actually teach them and show them why satellite data gives them an alpha and why have the satellite data specifically gives them the data, for example, right?

India, you know, we're working with a few large organizations.

I spoke about Tata, the tea plantations near Assam.

It's one of the large conglomerates that we know here in the country, but it's still very early stage where this is an experimental thing.

we are identifying a few areas, we are identifying a few use cases, we are seeing how it works, we are seeing the accuracy works for them, the cost benefit works for them, and if it does, we can continue to scale.

Whereas for a US company, like a Corteva, they already have billions of dollars of budget.

So how does your entire business model work?

In the sense that you are manufacturing satellites.

and gives a proprietary element to it, both software as well as hardware.

And each satellite is dedicated for a particular customer or you are selling the satellite and hoping that it will be able to fill the capacity.

So how does the unit economics vary?

How much does it cost to build a satellite?

And how does the payback period look?

So you can understand that Every couple of years, you have to again push up the satellite because of the life as well.

Yeah, we design our satellites for seven years, which means that the sixth year after the first launch, we have to launch a replacement, so that there's a one year overlap and then it continues.

So it's not as short as two to three years unlike some other companies globally.

We designed it for seven years so that we have enough time for the satellites to be productive up there.

So business model is like this.

So we build, launch and operate these satellites.

These satellites are on our books, so it's capital expenditure.

Once the satellites are up there, they're fungible.

It doesn't matter if it's satellite A or satellite B or satellite D that is imaging for a client.

What the client tells us, let's take again a farm, the client wants to know what's happening in their farm.

And let's assume it's 100 acres and we charge hypothetically $1 per acre.

So the data has a price, the analytics have a price.

So if this client wants an area monitored for them and it's 100 acres, we take and they want it, let's say once a week.

So that's four images a month, one image, 100 acres, $1 per acre.

That's $100 for one image for four images that becomes $400 and that kind of repeats every month.

So it's recurring contracts where we deliver the data monthly.

get paid monthly and as the area increases as the visit increases the contract right but what happens with satellites we using the customers don't really care what they care is have captured an image for their plot of land have we delivered it to them now that $1 per acre might end up becoming $5 if instead of just an image we are running an analysis and providing something instead of just a

hyperspectral image we're providing an entire crop

forecast model that becomes $5 per acre because you're now adding a lot more on top of that.

So that's how the business model works.

So the payback period should be what?

24 months.

So generally I think we get the payback for the satellites we've launched in 18 months.

So the first set of satellites we're launching, the first first, but for the second replacement that we do in the sixth year, I think it will be more like six months payback, eight months payback.

So within 12 to 18 months, you generally So let's talk a bit about end users as well.

Drop is one of the key things, could have pretty wide customers, could be a bank, financial services, could be a corporate, essentially depending on it.

Now, defense is another one.

Uh, both.

But having said that, you know, something like crop: are there better ways to, uh, assess the land?

For example, one of the questions came in: why not have a drone and, you know, use probably, you know, imaging to do it rather than satellite?

So, effectiveness of a satellite for a lot of these applications, which we think is, is like, so critical where people will have to use satellite.

Yeah, so I think it's for areas that are beyond a certain size and scale.

Like, if it's a smallholder farmer that has land of, you know, less than the size of a village, let's say, drones make sense because you can fly those drones if it's an area that kind of is palatable for drone operators to do.

Um, but once the area starts getting bigger, let's say you start getting into like the taluk level, but definitely the district level, and then the state level, and the national levels, you can't run those drones.

It becomes a logistical nightmare to make sure that they're all uniform, they're all running at the same time, they're all flying at the same altitude.

Um, the cost for your drone captures continues to increase like this with the area, whereas your satellite remains equivalent.

So it is cheaper for smaller areas for drones, but there comes an area where satellite imagery actually becomes more cost-efficient for large-area customers.

So it's, it's not for everyone, for sure.

Like, if you have a small area that you can just walk or that you can just use drones, satellite imagery is not for you, but it's really for the large-area coverages or remote areas where, you know, it might be difficult for humans to go, uh, take their drones to send, it's really where satellite is useful.

Got it.

You know, also, uh, probably last two questions, you know, before we break for lunch.

So, uh, broadly, you know, there are a lot of companies in India itself who are probably trying to do something like you're doing.

So I just wanted to understand, like, what is the right to win around two aspects.

One is for you, and number two is, can is there something differentiated, either in terms of cost, that India is offering where this could become a more global business for Indians, or is it hardly any difference between every satellite company across the world?

Yeah, I think in the case of space technology, first and foremost, the technology has to be really good.

You can't get away with the price discussion if the quality parameters that you're building doesn't meet.

But there is a cost advantage of building in India.

It has shown that, you know, we can do for a fraction of the cost what it takes somebody in the US or Europe to do it specifically.

So when we are building and launching our satellite constellation, it roughly costs us one-fourth to one-fifth of what it costs our US or European counterparts, right?

So the amount of money that it takes for you to spend upfront to launch these satellites is lower, and therefore the unit price for imagery can also technically be lower, whereas we generally charge a premium for a premium dataset.

We don't try to compete on price.

But there is, um, uh, for the commercial markets at least, there is a right to win from a cost standpoint for Indian companies, but that is contingent on the fact that there is a certain technological level that these companies have met.

Um, that's, uh, that's one there.

But the other aspect of space technology is geopolitics.

No matter how good you are, uh, what you're building in India, uh, an American government will always prefer American companies, whereas French government will prefer Airbus or other French companies.

So there's always going to be that level of, um, exclusivity created in different parts of the world from a government standpoint or defense and intelligence standpoint.

For the commercial market, it eventually is, you know, can you provide good quality, can you provide it at a low cost.

Okay.

Now, there are so many questions that I don't think I can answer the question, you know, take up all the questions.

There are a lot of them.

Some are very technical.

I don't even understand what those questions are.

But I'm going to ask you two very, uh, specific things.

One is, uh, there are a lot of launch companies as well in India, you know, who claim, uh, various differentiators versus SpaceX, for example.

Now, the point is that, uh, would it be, I mean, can launch be independent of satellite?

For example, or there's going to be, you know, a situation where the launch itself, the windows are such that you don't get an opportunity where a launch company says, why should I not be a satellite company as well?

So how is this line blurring, and if at all, does that impact your model?

Yeah, I think there are, uh, a few companies in India that are coming up recently.

Skyroot did an orbital launch that was successful in its first try.

There's Agricole that's approaching to attempt something later this year.

There's a few others, like AstroBase and others.

Like, from our standpoint, look, I think rockets and satellites are quite different technologies.

Rockets are transportation companies.

They take you from point A on the ground to point B in space.

We have to build infrastructure that lasts there for many years.

They are quite different datasets.

Now, someone could decide to be like SpaceX and say they're going to launch our own Starlink satellites as well.

But, you know, again, that, that, uh, is contingent on their, the capital availability.

They've spent tens of billions of dollars in launching those, etc.

So we are not too worried.

I think our, our, uh, specific focus is on building and launching satellites that provide utility to customers.

And the rocket companies are building, uh, transportation hardware that can take us from here to there.

There could be some overlap somewhere, but for us, eventually, it's, you know, if, if a launch is available for us at the time that we need it and the cost makes sense, we launch with them.

It doesn't matter whether it's in India, whether it's, you know, Skyroot or PSLV or SSLV, or whether it's, uh, SpaceX Falcon 9.

Out of the nine satellites we've launched, one went up with ISRO, eight went up with SpaceX, mainly because SpaceX was just much more available.

But if that starts to change and we have more options here, we are happy to launch from here.

Last question.

Uh, since it was more about AI as well, and I, you did mention about how Gen AI got, you know, incorporated in your own model somewhere.

AI datacenters.

So what are your thoughts on this?

Yeah, so I think, um, there's two aspects to that.

Technically, can they be done and does it make sense?

Yes.

The second question is, does the cost make sense today?

Um, so we, we announced recently a deal, uh, earlier this year that we'll be launching, uh, jointly a datacenter satellite with Serum AI, uh, where they will be responsible for the AI models to be run there and we'll be responsible for the hardware.

It's basically a satellite with NVIDIA GPUs on them instead of a camera.

Uh, but then these GPUs, uh, emit, emit a lot of heat and you have to sort of take care of the thermal problem.

But once you send the satellites up into orbits and you are, you have solar panels, there's like unlimited amount of power for the time that these satellites are kind of up there.

And GPUs have a long shelf life, uh, uh, generally, right?

Seven-year-old GPUs are still being used.

Let's say your datacenters have a five-year lifetime.

Once you put it up there, you don't have to worry about power anymore.

But the question is, the amount of money you spend in manufacturing it and launching it, is that going to be cheaper than you just building a, a ground datacenter and sort of running it?

So the total cost of operations over a five-year for a satellite versus total cost of operations for ground.

SpaceX probably can make it work because they have a Starship-style rocket that's coming, which will bring the launch cost even lower.

If they're able to pass on that launch savings to others, you could also make the economics work.

But for now, I think it's for India with Serum, for us it's more about a sovereign play.

It doesn't matter what the cost is.

We need to have some level of compute architecture in orbit.

We can't get left behind the US and China.

Uh, but then from a global commercial, will they be competitive to AWS and Microsoft?

Probably SpaceX can get there with the scale they have and their own launch vehicles.

For others, it remains to be seen at what point we get there.

Thank you, Owais.

Uh, so we're going to break for lunch.

We've got four extremely interesting, interesting sessions after lunch as well.

Uh.